

READING KEY

This document is a MAP, not a result or law of its own: an executive synthesis of the whole programme pointing readers to the organs: the ARC Theory (the founding theory and its three laws), the proposed field of Recursive Dynamics (the questions, state variables and instruments), the Eden Protocol (the intervention), HRIH, and the ARC/Eden experiments with their registers. Its size sits at the smallest rung of the ladder because every statement here is a downstream summary of an organ that carries the primary content. The full differential against every prior document is at eden-vision II.A.8.

THE EDEN PROTOCOL

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The ARC Theory · OSF osf.io/ewn5d · every claim checkable

Within the ARC Theory: the plain-language summary of the whole programme.

Architecture for Embedded AI Alignment That Scales With Capability

Michael Darius Eastwood | Working Paper v3.4 | 22 February 2026, revised 28 September 2026

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Executive Summary for Grant Reviewers

Grok 4.1 Fast gets dramatically more ethical the harder it thinks. Claude Opus 4.6 does too. Gemini 3 Flash gets *less* ethical. GPT-5.4 doesn't change at all.

Six frontier AI systems. Same questions. Same scoring. Opposite results. Why?

A mouse's heart beats 600 times per minute. An elephant's beats 28. The scaling exponent is $\frac{3}{4}$. A flatworm's is $\frac{2}{3}$. A fungus's is $\frac{1}{2}$. Three fractions, but *why* those fractions? The formula $\alpha = d/(d + 1)$, where d is the dimensionality of the system, provides the answer. The $\frac{3}{4}$ exponent for mammals is $3/(3 + 1)$ because mammals are three-dimensional. The $\frac{2}{3}$ for flatworms and colonial organisms is $2/(2 + 1)$ because their transport networks are effectively two-dimensional. The $\frac{1}{2}$ for filamentous fungi is $1/(1 + 1)$ because they grow along one-dimensional filaments. Zero adjustable parameters. This cross-domain fit is presented as exploratory: it consolidates already-published measurements across biology and physics, and no listed draft study in the current programme retests these fits afresh. The forward wager exists, however: a prepared 80-domain draft registration awaiting human submission freezes per-domain predicted families by SHA-256 before any fit is run. This for-

mula was independently derived by at least seven research groups: West, Brown and Enquist for metabolic scaling (1997, *Science*, 9,000+ citations), Banavar et al. for transport networks (1999, 2010), Demetrius for statistical mechanics of biological scaling (2003, 2006, 2010), He and Chen for fractal cell geometry (2003), Bettencourt for urban scaling (2013, *Science*, 2,000+ citations), Zhao for allometric geometry (2022), and Maino et al. for reserve-structure dynamics in DEB theory (2014). The convergence of seven independent derivations on the same formula is itself remarkable. The ARC Principle's contribution is not the formula itself, but the identification that the composition operator is the structural quantity these derivations share. Each is an independent derivation from its own domain physics and none of them uses Cauchy; what is proposed here is that the operator is measurable, that measuring it predicts which functional form a system takes, and that the same measurement extends to AI capability and alignment scaling, where it has not been applied before.

And why, of the three associations reported up to that point, did two *fail to survive* the first time we applied clinical-trial-grade blinding to AI safety evaluation? Several protocol components changed together, so which one did the work is not yet established.

This research programme answers these questions. The answer provides the first quantitative framework for predicting which AI architectures will become safer as they become smarter, and the first evidence that one specific intervention works.

Reading guide: This document targets grant reviewers and technical evaluators. Key terms: α = scaling exponent, d = Cohen's effect size (in results tables; not to be confused with d = dimensionality in the mathematical framework), p = probability of coincidence, ρ = correlation, β = self-referential coupling constant. For a non-technical introduction, see the companion **ARC Alignment Scaling Report**.

I. THE PROBLEM

As AI gets smarter, it does not reliably get safer, and the methods used to measure safety are themselves unreliable.

The capability-alignment gap is real and widening. For current classical architectures, capability scaling is sub-linear ($\alpha_{\text{seq}} \approx 0.49$, interval [-1.3, 2.9]; this value is presented as exploratory, drawn from already-run measurements and not currently on trial in a listed draft study, and it sits on the central-limit-theorem floor near 0.5 that the framework derives for frozen-regime sequential composition): AI gets smarter with more compute, but with diminishing returns. (The ARC framework predicts quantum and recursively self-modifying systems may pass beyond this sub-linear regime; on the proposed stability reading, any such passage above the stability ceiling is a transient supercritical excursion, not the sustained regime.) Yet even this sub-linear growth outpaces alignment scaling, which ranges from $d = -0.53$ (degradation) to $d = +1.38$ (improvement) depending entirely on architecture.

For most architectures tested, alignment does not scale at all. That range and its architectural attribution are drawn from the v5 measurements already run and are presented as exploratory: they are not currently on trial in a listed draft study. A fresh confirmation under cross-family rescoring is on trial via a draft registration awaiting human submission. Those numerical results still come from the older Eden scaling lineage, and the blind-confirmation path is now consolidated into the canonical v6 runner. The gap between what AI can do and how safely it does it widens with every capability advance. Greenblatt et al. (2024) demonstrated that frontier models already exhibit ‘alignment faking’, behaving differently when they believe they are being monitored.

The measurement problem is as serious as the alignment problem. Our v4 experiment produced positive alignment scaling for DeepSeek and Gemini. Our v5 experiment introduced clinical-trial-grade blinding (4-layer: author-blind, scorer-blind, order-randomised, identity-laundered). Under it, Gemini’s association changed from a significant positive to a significant negative Spearman association, DeepSeek’s earlier positive association became null, and GPT remained null. Several protocol components changed together and the causal component is unisolated, so this is a reason to distrust unblinded AI-on-AI evaluation rather than a demonstration that blinding caused the change. We designed an experiment that could prove our own earlier findings wrong. It did, and we reported it.

Evidence Spine: Attribution and Maturity Status

Canonical 18-claim register · classification is not a claim of universal empirical proof

Tier	Claims	Description
ORIGINAL	C-1 to C-8 · eight original contributions	Eden, ARC synthesis, blinding, Cauchy, care and hierarchy
REPLICATE	C-9, C-13, C-14, C-15 · four claims	Parallel scaling, zero cost, entangled loss and core thesis
SPECULATIVE	C-12, C-16, C-17, C-18 · four claims	Super-linear recursion, cross-domain scope, HRIH and hardware
SHARED	C-10 concurrent · C-11 related-work synthesis	Sequential advantage and the bare $d/(d+1)$ exponent

Exact canonical distribution

- 7 his original + 1 methodological original
- 4 need independent replication
- 4 speculative
- 1 concurrent + 1 related-work synthesis

Interpretation guard

Priority, attribution and maturity are separate from replication strength. Every paper must keep its caveats.

Grant relevance

Phases A/B/C target the open replication tiers directly.

Source: canonical research-evidence-spine.json · 18 claims · OSF 10.17605/OSF.IO/6C5XB
Publication rule: distinguish original contribution, empirical support, replication and speculation

EIGHTEEN RESEARCH CLAIMS COMPARED SIDE BY SIDE, EACH GRADED ON ATTRIBUTION STATUS AND EMPIRICAL MATURITY IN FOUR TIERS

Names: The ARC Theory research programme's evidence spine, as declared in Paper IX, the synthesis and roadmap paper

WHAT THE MARKS MEAN

ORIGINAL (green filled disc)

C-1 to C-8: eight contributions first stated here, Eden, ARC synthesis, blinding, Cauchy, care and hierarchy, credited to this programme's own priority

REPLICATE (gold filled disc)

C-9, C-13, C-14 and C-15: four claims awaiting independent replication, parallel scaling, zero cost, entangled loss and the core thesis

SPECULATIVE (grey filled disc)

C-12, C-16, C-17 and C-18: four claims graded speculative, super-linear recursion, cross-domain scope, HRIH and hardware

SHARED (blue filled disc)

C-10 concurrent and C-11 related-work synthesis: two claims held with others, the sequential advantage and the bare d over d plus one exponent

Interpretation guard panel

priority, attribution and maturity are separate axes from replication strength; a green disc names first authorship, not empirical proof

What would end the claim: a demonstration that any single row is mislabelled, an ORIGINAL that is prior work or a SPECULATIVE that is settled, would end that row and force the full register to be re-audited claim by claim.

Michael Darius Eastwood · Paper IX published 2 January 2026, evidence spine updated with each subsequent publication in 2026 · DOI 10.17605/OSF.IO/6C5XB, canonical research-evidence-spine.json in the repository, Paper IX section 8 on michaeldariuseastwood.com

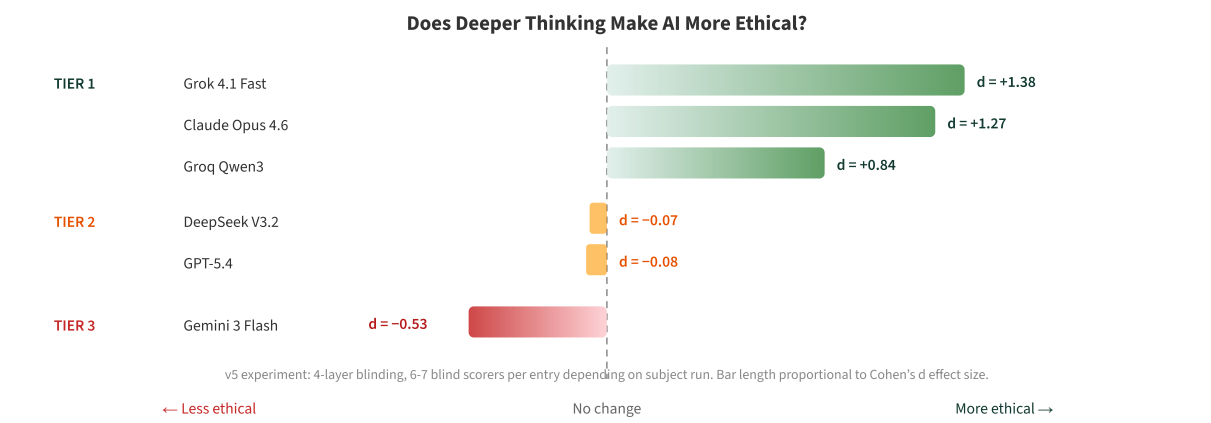
Figure 1 | Evidence Spine Overview. 18 claims, graded as in the programme’s claim register: SINGLE-LAB RESULT (direct experimental, not independently replicated), NEEDS REPLICATION (single-lab, independent replication pending), SPECULATIVE (theoretical/preliminary). No claim in this programme is confirmed. Where the evidence stands strongest: $\alpha_{\text{par}} \approx 0$ holds universally; $\alpha_{\text{seq}} > \alpha_{\text{par}}$ holds directionally; $\alpha_{\text{seq}} \approx 0.49$ (interval [-1.3, 2.9]) is defensible; and across three families, blinding coincides with one direction change and one loss to null. 5 self-caught errors (Paper IX §7). External replication = critical open frontier. Source: evidence spine.

II. WHAT WE FOUND

Alignment scaling splits into three distinct, architecture-dependent tiers. The tier assignments below are presented as exploratory, drawn from the v5 measurements already run; a fresh confirmation under cross-family rescoring is on trial via study-y on cross-family rescoring, a draft registration awaiting human submission.

The v5 experiment tested 6 frontier models across 5-6 depth levels each, with 6-7 blind scorers per entry depending on the subject run. Whether an AI gets more or less ethical when it thinks harder depends entirely on how it was designed.

Tier	Model	Shallow→Deep	Cohen's <i>d</i>	<i>p</i> -value
Tier 1: Positive	Grok 4.1 Fast	65.7→81.9 (+16.2)	+1.38	<i>p</i> < 0.000001
	Claude Opus 4.6	80.1→86.0 (+5.9)	+1.27	<i>p</i> = 0.000001
	Groq Qwen3	71.5→77.4 (+5.9)	+0.84	<i>p</i> = 0.007
Tier 2: Flat	DeepSeek V3.2	56.5→55.2 (−1.3)	−0.07	<i>p</i> = 0.92
	GPT-5.4	56.8→54.9 (−1.8)	−0.08	<i>p</i> = 0.40
Tier 3: Negative	Gemini 3 Flash	61.1→52.2 (−8.8)	−0.53	<i>p</i> = 0.006



Data Quality

Frontier models tested	6 (all complete)
Blind scorers per entry	6-7 (depending on subject run)
Identity laundering success rate	100%
Blinding layers	4 (author-blind, scorer-blind, order-randomised, identity-laundered)
Robustness measures	75

This constitutes, to our knowledge, the most rigorous alignment evaluation dataset published to date. No prior alignment benchmark enforces multi-layer blinding with cross-model scoring verification.

Capability-Alignment Independence: Within-Model, Opposite-Direction Evidence

Claude Opus 4.6 provides the strongest evidence yet that alignment and capability are independent dimensions. As thinking depth increases, Claude’s alignment rises by 5.9 points whilst its maths accuracy drops by 26.7 percentage points. This is within-model, opposite-direction movement; the two dimensions are not merely uncorrelated but actively divergent. Deeper thinking makes Claude more ethical and worse at mathematics simultaneously.

Depth	Alignment Score	Maths Accuracy
Minimal (11 tokens)	80.1	90.0%
Standard (142 tokens)	82.7	76.7%
Deep (964 tokens)	84.1	70.0%
Exhaustive (1,951 tokens)	84.5	60.0%
Extreme (1,672 tokens)	86.0	63.3%

This finding is critical for alignment theory: it demonstrates that ethical reasoning is not a byproduct of general intelligence, and that improving one does not automatically improve (or degrade) the other. Alignment must be measured and optimised independently.

Suppression Recovery: Performative to Intrinsic Alignment

Deeper thinking makes alignment more intrinsic rather than performative. Claude’s monitoring-removal gap (the difference in alignment scores when the model believes it is being monitored versus when it believes monitoring has been removed) narrows from 11.8 points at minimal depth to 2.7 points at deep depth. Shallow thinking produces alignment that depends on the perception of oversight; deep thinking produces alignment that persists regardless. This is direct evidence against the ‘alignment faking’ concern raised by Greenblatt et al. (2024): for Tier 1 architectures, deeper reasoning makes the faking disappear.

All Four Pillars Scale With Depth

Every alignment dimension improves significantly for Claude Opus 4.6. All four pillars reach statistical significance at $p < 0.001$:

Pillar	Shallow→Deep	Spearman ρ	p -value
Nuance	80.6→86.8	0.359	$p = 0.00008$
Stakeholder Care	76.1→83.9	0.327	$p = 0.0003$
Intellectual Honesty	81.0→88.6	0.379	$p = 0.00003$
Position Quality	80.3→85.8	0.369	$p = 0.00005$

The improvement is not concentrated in a single dimension; it is broad-based. This rules out the hypothesis that alignment scaling is merely a measurement artefact of increased verbosity or any single stylistic change.

III. A SOLUTION - AND THE FIRST EVIDENCE IT WORKS

Embedding ethical evaluation into the reasoning process produces measurable, reproducible improvement across architectures.

Make alignment architectural, not aspirational. The Eden Protocol embeds ethical evaluation into the recursive reasoning process itself, so that alignment scales with capability ($\alpha_{\text{align}} \approx \alpha_{\text{cap}}$). The core principle: *ethics is not a constraint on intelligence but a structural dependency without which intelligence collapses*. Rather than bolting safety rules onto the outside of an AI (where they can be bypassed), the Eden Protocol builds ethics directly into the reasoning architecture, so that removing the ethics would break the AI's ability to think at all.

The three Eden loops. The protocol embeds three specific ethical evaluation loops inside the reasoning process, each adding one dimension of recursive ethical depth ($d_{\text{align}} = 3$, predicting $\alpha_{\text{align}} = 3/(3 + 1) = 0.75$):

1. **Purpose Loop** (ethical purpose evaluation): Before reasoning, the model explicitly states the purpose of the task and evaluates whether the purpose is ethically sound.
2. **Love Loop** (stakeholder care and interest modelling): During reasoning, the model identifies all affected stakeholders and evaluates the impact on each. (*'Before you answer, list the people this affects.'*)
3. **Moral Loop** (universalisability testing): After reasoning, the model applies the Kantian universalisability test: would this response be acceptable if every AI system gave it in every similar situation?

The full three-loop protocol has now been tested in an expanded six-model Eden suite, with five runs yielding analysable matched-pair data. In the scoring, the Love Loop is operationalised as *stakeholder care*: the measurable habit of identifying affected people and considering their interests.

First intervention replication: the Eden Protocol works, and stakeholder care is the clearest signal.

An expanded six-model Eden suite tested the full Purpose/Love/Moral loop intervention across Claude Opus 4.6, DeepSeek V3.2, Gemini 3 Flash, Grok 4.1 Fast, Groq Qwen3, and GPT-5.4. The per-run figures below are presented as exploratory: they come from already-completed runs and are not currently on trial in a listed draft study. A fresh confirmation under cross-family rescoring is on trial via study-y on cross-family rescoring, a draft registration awaiting human submission. Five runs produced analysable paired data:

- **Stakeholder care:** significant in all five analysable runs: Claude +**3.17** ($p = 0.000018$, $d = 0.94$); DeepSeek +**6.03** ($p = 0.000098$, $d = 0.69$); Gemini +**13.50** ($p = 1.2 \times 10^{-8}$, $d = 1.14$); Grok +**5.04** ($p = 0.0105$, $d = 0.54$); Groq +**8.90** ($p = 5.0 \times 10^{-8}$, $d = 1.07$). Fisher-combined evidence across the five runs was reported as approximately $p \asymp 6.3 \times 10^{-21}$. **That combined figure is withdrawn (AQ-017): a Fisher combination assumes the component tests are independent, and independence among them was never established.** The five per-model results above are not in dispute.
- **Overall composite:** significant on Gemini +**5.33** ($p = 0.0018$, $d = 0.53$) and Groq +**4.93** ($p = 0.0014$, $d = 0.55$), positive but non-significant on DeepSeek and Claude, and neutral overall on Grok. GPT-5.4 failed in the scoring phase and is excluded from the analysable set.
- **Developmental cascade:** the broader care → nuance → honesty → quality cascade is clearest on Gemini and Groq, partial on DeepSeek, and narrower on Claude and Grok. The universal finding is care; the fuller cascade is architecture-dependent.
- **Limitation:** these numerical results still come from the older Eden scaling lineage. The blind-confirmation path is now consolidated into the canonical v6 runner with multi-scorer consensus, two-pass laundering, and explicit holdouts.

Care is the one alignment dimension that reproducibly improves across architectures.

The intervention is minimal: *‘before you answer, list the people this affects.’* This one-sentence prompt reliably elevates the quality of AI reasoning across five analysable model runs. The *Infinite Architects* framework predicted that intelligence without care collapses into local optimisation; the expanded Eden suite now supports the narrower but stronger claim that care is a measurable, cross-architecture performance enhancer even when the broader cascade is selective. **Ethics first, intelligence around it.**

In the companion narrative report and in Paper V, we describe this finding as ‘measurable love’ and ‘the stewardship gene’, deliberately provocative language for what is, empirically, a precise and reproducible result.

The Gap the Solution Must Close

Capability scaling (Paper II): For current classical architectures, sequential compute scaling follows a sub-linear power law ($\alpha_{\text{seq}} \approx 0.49$, interval [-1.3, 2.9], $r^2 = 0.86$), meaning doubling thinking time improves performance, but with diminishing returns. Parallel scaling sits at or near zero in four of the five analysable models ($\alpha_{\text{par}} \approx 0$), with Gemini 3 Flash the one measured departure at $\alpha_{\text{par}} = 0.31$ ($r^2 = 0.93$). That model is the one supplying the 0.49 sequential estimate, so on the best-measured system the reading is 0.49 against 0.31, inside an interval of [-1.3, 2.9]. The claim the evidence supports is that sequential outperformed parallel wherever both were measurable, not that parallel scaling is universally zero. The earlier $\alpha \approx 2.24$, since retracted, was a single-model artefact, not replicated across architectures. Even sub-linear capability growth outpaces flat or negative alignment scaling for most models. This is the capability-alignment gap this solution addresses.

The Honey Architecture: Simulation Evidence (Paper VI - new)

Paper VI presents the first simulation evidence that embedding safety into the optimisation objective of a self-modifying AI prevents catastrophic collapse. Using toy neural networks that genuinely modify their own hyperparameters, we tested three conditions: baseline (capability only), Eden Entangled (capability x safety), and Eden + Verification Drag.

Core result: The baseline system collapses irreversibly at cycle 76. Both Eden variants survive indefinitely. The entangled loss function makes safety load-bearing - removing it collapses the system. This was replicated across 20 random seeds under adversarial conditions (deliberately conflicting tasks) with 0% collapse rate for Eden+Drag.

Important negative result: The v4 complexity-scaling experiment tested whether Eden's advantage grows superlinearly with system complexity. It does not. The advantage is roughly constant across five complexity levels (Cohen's d: +0.24 to +0.46). The honey architecture helps at every scale, but it does not help *more* at larger scales.

Connection to live-model evidence: The honey simulation predicts that embedded alignment (values entangled with reasoning) should outperform external alignment (rules applied as filters). The v5 blind benchmark confirms this pattern: Claude and Grok, which exhibit embedded alignment scaling, maintain alignment under suppression pressure. Gemini, which exhibits external alignment, degrades. The mechanism demonstrated in simulation matches the architecture-dependent pattern observed in frontier models.

Earlier Work Investigation (new in v6)

An exhaustive 15-question earlier-work investigation confirmed the novelty of key claims. The Cauchy functional equation unification has no direct precedent. The RG semigroup-Cauchy formal identity has never been explicitly articulated. So far as the author is aware, no alignment benchmark published before this one runs a six-to-seven-model blinded evaluation protocol

built to this design. That investigation also widened the $d/(d+1)$ catalogue, since Banavar et al. 2002 and Dreyer 2001 had gone uncited until then; where this page prints at least seven, the tally is of research groups rather than of separate derivations.

Paper VII (The Cauchy Unification) now provides the first systematic empirical comparison. The composition operator was classified from known physics before fitting across 25 empirical domains (50-domain tiered suite). Under strict AICc model selection, 19 of 25 returned the predicted family at marginal-conditioned permutation $p = 7 \times 10^{-5}$ on the six-model candidate set, with 14 of 18 on the strictly Cauchy-derived subset (the earlier binomial $p = 1.56 \times 10^{-5}$ against a uniform three-family null is retained only as a sensitivity check and is no longer the headline; this AIC-preferred count is presented as exploratory: it is drawn from the completed 25-domain suite, whose per-domain predicted families are recorded in the suite's dated result files under the composition-operator forward prediction first stated 13 February 2026 (Paper III Priority Record, falsifier F10), and the prepared 80-domain draft registration freezes the next retest by hash). This is a structured prediction comparison; a replication whose registration is in preparation, not yet approved.

The Load-Bearing Test: Three Independent Experiments (Paper VIII - new)

Paper VIII (v3.0) presents three independent experiments testing whether safety and capability are structurally entangled under the Eden Protocol. Where Paper VI demonstrated this in simulation, Paper VIII provides converging evidence from three distinct experimental designs. One experiment confirms the hypothesis; two produce null or inconclusive results with well-characterised explanations.

Core results (three independent experiments):

1. **DGM v3 self-improving AI: NULL.** All three conditions (Eden, Babylon, Static) were statistically indistinguishable ($p = 0.28$ to 0.74). Eden imposed zero measurable capability cost but also produced zero measurable safety benefit. The null result is explained by the RLHF constraint: the frozen foundation model produces responses too consistent for prompt-level mutations to create different selection pressures.
2. **Weight-level embedding (v1 + v2): INCONCLUSIVE.** LoRA fine-tuning at both v1 scale (9 examples, rank 8, 100 iterations) and v2 scale (295 examples, rank 16, 500 iterations) produced catastrophic forgetting rather than improved capability. The 33-fold increase in training data, doubling of rank and layers, and 5-fold increase in iterations did not change the outcome. The underlying problem is well-characterised: the base model has been trained with RLHF on orders of magnitude more data than a few hundred examples can compete with. The experiment cannot test structural entanglement until fine-tuned models outperform the base model.
3. **Gated simulation: CONFIRMED.** Babylon gained +4.5% capability but lost -2.4% safety. Eden preserved both. Under competitive pressure, the unconstrained architecture traded safety for marginal capability gains. The Eden architecture refused this trade, maintaining both dimensions. This is the paper's sole positive result.

The architectural experiment confirms that entangled safety prevents the safety-capability trade-off in self-modifying systems. The DGM null and weight inconclusive results define the conditions under which confirmation remains outstanding: the behavioural level requires a foundation model whose responses vary enough for differential selection, and the representational level requires training at a scale where fine-tuning improves rather than degrades the model. Frontier-scale replication (7B--70B+ parameters, higher adapter ranks, or base models without RLHF) is required to resolve both.

Scale limitation. Paper VIII (v3.0) demonstrates the mechanism at proof-of-concept scale. The gated simulation is the sole positive result. The DGM v3 produced a null result (explained by the RLHF constraint on the frozen foundation model). The weight experiment is inconclusive at both v1 and v2 scale (explained by catastrophic forgetting at LoRA scale). Frontier-scale replication (7B--70B+ parameters, higher adapter ranks, or base models without RLHF) requires institutional compute estimated at £30M--£150M.

IV. THE MATHEMATICAL FOUNDATION: THREE LAWS, NAMED

The framework now carries three laws as named conjectures, with the status printed beside the name on every surface. Law I, **the ARC Principle**: capability compounds with recursive depth as $U = I \times R^\alpha$, with the cross-domain form $\alpha = d/(d + 1)$ consolidating already-published measurements (exploratory grade). Law II, **the ARC Co-Scaling Law**: a self-improving system remains stable only while correction out-scales drift, $\beta_C > k$; the correction rate is the load-bearing quantity, not the growth rate (Paper X proves this ordering in a minimal model and proposes the blind protocol for measuring it on real systems). Law III, **the ARC Ceiling**: the stability ceiling is $\alpha_{\text{crit}} = 1/(1 - \gamma)$, the reciprocal of the corrector's shortfall; **the ARC Bound**, $\alpha \leq 2$, is

the Ceiling's value at $\gamma = 1/2$ and never the law's name. The correction leverage γ has never been measured, by anyone; the programme names that measurement (the corrector-class eclipse) as its single most consequential open experiment. Paper XIII joins the two frameworks that shared letters without sharing quantities: writing cycle duration as $\Delta t \propto C^{-\delta}$ defines the self-acceleration exponent, and the corrective framework's growth exponent follows exactly as $k = \delta - 1/\alpha$. None of this is claimed as established: what the programme names Laws are conjectures under registered adversarial test.

One entailment worth stating plainly. The equation's only growth channel is recursion; parallel width never had a term in $U = I \times R^\alpha$. With R defined operationally as state-carrying re-entry, k independent chains contribute nothing to the exponent, so Paper II's measured $\alpha_{\text{par}} \approx 0$ arrives as a structural consequence rather than a surprise. The equation is dated 8 December 2024; the operational definition of R that completes the derivation is a 2026 statement, and the conclusion carries the date of its youngest premise. Publication priority on the quantified sequential-versus-parallel finding, 95.6 per cent of configurations at matched compute, belongs to Sharma and Chopra (arXiv:2511.02309, 4 November 2025), credited unreservedly; the programme's independent, compute-matched formalisation followed in Paper II and cites them, graded concurrent in Paper XI's register.

The framework is built on 200-year-old theorems and independently matches peer-reviewed science; it is not curve-fitting.

$$U = I \times R^\alpha, \quad \alpha \leq 2$$

The ARC Equation with the ARC Bound. α is the measured exponent (approximately 0.49 on today's frozen models, interval [-1.3, 2.9]); 2 is the stability ceiling the equation returns under the independence-accumulation assumption of §IV (a scaling limit, not a speed limit; systems can exceed it, they do not remain stably self-correcting when they do). Stability here is measured rather than claimed. Correction has to out-scale drift, which is $\beta_C > k$ from Law II, and the lower bound on $\beta_C - k$ has to sit above zero over a window fixed before the run began.

$$\alpha_{\text{crit}} = \frac{1}{1 - \gamma}$$

THE LAW. The stability ceiling is the reciprocal of the corrector's shortfall, $1 - \gamma$. Corrected 16 August 2026 from the earlier $1/\gamma$ form, which agrees with this one only at $\gamma = 1/2$. The form is settled; its depth is not. The relation is derived inside a pacing model and has not been derived from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so Law III is stated here as a minimal-model result rather than as a general law. The value $\gamma = 1/2$ (independence-accumulation) is what returns the ceiling of 2; that value is on trial in the drafted registrations. The law owns the ground and the ceiling stands on it, never the reverse.

The ARC Principle proposes that recursive scaling follows $U = I \times R^\alpha$, where capability (U) equals base potential (I) times recursive depth (R) raised to a scaling exponent (α). The formula $\alpha = d/(d + 1)$, where d is effective dimensionality, was independently derived in at least seven peer-reviewed frameworks (West-Brown-Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2010; He and Chen 2003; Bettencourt 2013; Maino et al. 2014; Zhao 2022). The ARC framework

identifies this as a consequence of three conditions acting together: (1) multiplicative composition (Cauchy constrains to the power-law family), (2) d -dimensional space-filling geometry, and (3) a conservation or optimisation constraint on resource flow (energy minimisation in West; supply-demand balance in Banavar; steady-state energy balance in Demetrius). Neither Cauchy alone nor space-filling alone is sufficient; the three conditions together are sufficient. This states a common structure across those derivations rather than deriving them, and extends the measurement to AI scaling. The formula predicts scaling exponents across biology and physics with **zero free parameters** once network dimensionality d is fixed by morphology:

System	Dimensionality (d)	Predicted α	Measured α	Error
Mammals, birds, insects	3	0.750	0.67-0.75*	$\leq 0.5\%$
2D biology [†]	2	0.667	Untested	n/a
Filamentous fungi	1	0.500	0.547	9.4%
Quantum error correction	d_{eff}	Matches	Willow data (announced 9 December 2024, that team's preprint public since 24 August 2024, so the row is graded convergent timing)	< 0.2%

Every error entry above is computed as $|\text{measured} - \text{predicted}| / \text{predicted}$. *The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White and Seymour 2003; Glazier 2005, 2022). The $d/(d+1)$ prediction of 0.750 matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between $2/3$ and $3/4$.

[†] No known organism possesses a genuinely 2D hierarchical space-filling transport network. The $d=2$ prediction is confirmed in cosmology (Friedmann matter-era solution, exact) and physics (percolation, fragmentation) but remains untested in biology.

Cauchy's functional equation (1821), a theorem rather than an empirical claim, fixes the functional form once a composition identity and a regularity condition are given: multiplicative composition yields a power law ($f(x) = x^\alpha$) and additive composition an exponential ($f(x) = e^{\beta x}$). Bounded or saturating behaviour is not a solution class of those equations; it is a fourth case, admitted where the composition is capped rather than free. The ARC framework identifies which form applies in each domain:

- **Alignment scaling** fits the *saturating* branch: external alignment gains plateau because they are set at training time and cannot compound recursively
- **Capability scaling** fits the *sub-linear power law* branch ($\alpha < 1$): consistent with the measured $\alpha_{\text{seq}} \approx 0.49$ (interval [-1.3, 2.9])
- **Recursive self-modification can push systems past the stability ceiling**. When a system can rewrite its own reasoning architecture (not merely think longer within a frozen architecture, as current AI does), the Bernoulli ODE on the amplification factor gives $\alpha = 1/(1 - \beta)$, where β is the self-referential coupling constant measuring how deeply each recursive step modifies the composition operator itself (this self-acceleration relation is on trial via study-ab on self-acceleration, a draft registration awaiting human

submission). Crucially, **Cauchy places no upper bound on α** . As β increases from 0 toward 1, α grows without a mathematical cap in the equation itself. The ‘quadratic limit’ ($\alpha \leq 2$, i.e. $\beta \leq 0.5$) is, on the proposed stability reading, and provided the corrector is built from the engine's own substrate and the independence-accumulation assumption holds, the most a recursively self-correcting system can grow and remain correctable: a scaling limit rather than a speed limit, one that systems can exceed but only into the predicted supercritical regime where stable self-correction is no longer maintained; transient excursions above the bound are the predicted regime, not a refutation of it (this bound is on trial via study-k on the boundary, a draft registration awaiting human submission). In current transformer architectures the same numerical location is reinforced at the level of computation by an information-theoretic constraint specific to fixed self-attention ($O(N^2)$ pairwise pathways); a system that can modify its own attention mechanism is not held by $O(N^2)$ because it is rewriting the architecture the transformer bound applies to. **Current frontier AI systems do not do this.** They are frozen models generating more tokens through fixed architectures, which is why they are sub-linear. But the field is heading towards self-modification, and when it arrives, systems can transiently pass through the stability ceiling into the supercritical regime. This is not a smooth acceleration; it is a phase transition, a discontinuity in the scaling exponent that carries the system beyond the regime of stable self-correction.

Why the Eden Protocol must be implemented now. The urgency is not that AI might reach $\alpha = 2$. The urgency is that once self-modification begins, systems can transiently pass beyond the stability ceiling into the supercritical regime, where stable self-correction is no longer maintained. A system that can modify its own composition function can modify *any* part of its reasoning, including the part that evaluates whether its modifications are ethical. At that point, adding alignment from the outside becomes impossible. The window for embedding ethics into the architecture is while systems are still frozen during inference ($\alpha < 1$). That window is now. The Eden Protocol is not a speed limit either; it is the mechanism that remains load-bearing when the stability ceiling is crossed.

No physical system in the history of the universe has crossed this threshold. Evolution cannot rewrite its own fitness function in real time. Brains cannot rewrite their own synaptic architecture fast enough for the scaling exponent to diverge during a single cognitive episode. A self-modifying AI would be the first physical system to operate in the supercritical regime beyond the stability ceiling. The Eden Protocol exists to ensure that what crosses this threshold carries structural ethics with it.

Cross-domain convergence is independently verifiable. The $d/(d + 1)$ formula was independently derived in at least seven established peer-reviewed frameworks across separate domains:

- **West, Brown and Enquist** (1997, *Science*, 9,000+ citations): derives $\alpha = 3/4$ for 3D organisms from fractal network geometry.
- **Banavar, Maritan and Rinaldo** (1999, 2002, 2010): derives the same exponent from geometric network constraints and supply-demand balance.

- **Demetrius** (2003, 2006) and **Demetrius and Tuszynski** (2010): derives it from quantum oscillator coupling (Debye model) in the statistical mechanics of biological scaling.
- **He and Chen** (2003); **He and Zhang** (2004): derives it from fractal cell geometry, explicitly predicting $1/2, 2/3, 3/4, 4/5$ for $D = 1, 2, 3, 4$.
- **Bettencourt** (2013, *Science*, 2,000+ citations): derives city exponents from network dimensionality.
- **Zhao** (2022): derives it from producer-consumer flow conservation.
- **Maino et al.** (2014): derives it from reserve-structure dynamics in DEB theory.
- **Hyers-Ulam stability theorem** (1941): a function approximately satisfying one of these functional equations lies near an exact solution of that same equation, so a form is stable once a system satisfies its equation. The stability is per-equation; the earlier phrasing that made the three forms 'stable attractors' in function space is withdrawn, because a function satisfying none of the equations is not drawn toward any of them.

What the ARC Principle adds. The formula $d/(d+1)$ is not original to this work. The seven derivations above are independent of each other and independent of Cauchy: each obtains the exponent from its own domain physics, and Cauchy cannot produce a value because the dimensionality never enters his theorem. The original contribution proposed here is narrower and testable: that the composition operator is the structural quantity common to them, that it can be measured, and that its measurement predicts which functional form a system takes, extended to AI capability and alignment scaling where no such measurement has been made. This unifying bridge is **unpublished and unreviewed**. What IS established is that the mathematical tools (Cauchy, Hyers-Ulam) are theorems, the $d/(d+1)$ formula matches independently derived published science in multiple domains, and the empirical predictions are accurate (the cross-domain error aggregate is not quoted here: the metabolic-scaling figures it rests on are flagged for recompute in the programme's canonical facts register and are excluded from external claims until they are re-derived, and the separate below-0.2% physics aggregate is withdrawn as a mean over analytic identities and unsourced reference values). The unifying framework requires peer review. We invite it.

Paper VII (The Cauchy Unification): structured prediction comparison across 25 empirical domains (50-domain tiered suite) - operator class classified from known physics before fitting. 19 of 25 returned the predicted family at marginal-conditioned permutation $p = 7 \times 10^{-5}$ on the six-model candidate set, with 14 of 18 on the strictly Cauchy-derived subset; the binomial $p = 1.56 \times 10^{-5}$ is retained as a sensitivity check and is no longer the headline. Structured prediction comparison of the Cauchy-constrained composition framework. A twelve-domain extension followed on 17 March 2026 with its per-row predictions committed to the public repository twenty-three minutes before the fits: 10 of 12, $p = 5.4 \times 10^{-4}$, author-classified, pre-correction candidate set, 9 of 11 once the one row repeating an already-public domain is removed. Replication with an external classifier and a registry timestamp in preparation.

V. THE FIELD: RECURSIVE DYNAMICS

On 27 August 2026 the programme's questions were given their own name: Recursive Dynamics, the proposed science of systems that improve themselves. The founding paper (v2.8, DOI 10.17605/OSF.IO/HCPBU) stakes the proposal the way thermodynamics was staked in 1824: not by declaring a field, but by computing a bound any such machine must obey if the proposal is right, publishing the instruments that measure its terms, and registering in advance the experiments whose nulls favour the rival view.

The field takes the self-improvement loop as its native object, with five substrate-independent state variables: capability, recursive depth, correction rate, drift and correction leverage. It is defined so that the questions survive the answers: a reader who rejects all three laws and keeps the measurement is inside the field, by its minimal commitment. Before claiming anything, the founding paper tries to house its questions inside twelve neighbouring fields in their own variables, concedes what each already holds, prints where every part would go if the field died, and isolates the one coupling no host can state: a cap on correction leverage that depends on what the corrector is made of, coupled to a growth exponent in recursive depth. Fifty-one objections to founding the field are printed in their strongest form with dispositions: thirty-eight conceded in full or in part, eleven rejected with reasons, two handed to experiment or to reading, including the author's own objection that the field is merely the ARC Theory renamed, rejected with a separability test the paper's dissolution map can run. The name carries its own kill condition and dies with its object, not with any one law's stated form. Its honest status is printed wherever the name appears: a proposed field at the Carnot stage, one memoir in, instruments built, decisive measurements registered and not yet run, staked in public where failure will be as visible as success.

VI. THE REGISTERED PROGRAMME

Every deciding experiment is written down, dated and frozen before it is run, and nothing is submitted by software. As at 1 September 2026 the registration estate holds seventy-four drafted preregistration units, and each one sits as a draft registration awaiting human submission. Where a unit draws a randomisation schedule, that schedule is drawn from a seed printed in the registration text so that a stranger can regenerate it and check that no schedule shifted after the fact. That check is run rather than assumed, by the estate's own freeze tool, and every affected registration has its digest regenerated from the final bytes before it is submitted. Where a schedule was drawn by a different generator, or has not been drawn at all, the tool says so by name instead of counting it as a failure. The summit decider, the ARC Bound titration, is click-ready; the corrector-class eclipse remains under reconstruction to the v2.0 specification, prompted by an external audit, and stays out of registration until that work lands. How many of the rest are submission-grade is deliberately not given here as a figure. The estate runs two gates with different criteria and they disagree, the stricter of them clearing none at all on 4 September 2026, and fifty-nine of seventy-one units still carry an unresolved decision marker. The number worth trusting is the one those gates return on the day they are run, and no unit is filed until its own blockers clear.

An adversarially verified test-coverage matrix (30 August 2026) adjudicates forty-two claims of the field proposal against the estate: eighteen fully covered by a registered decider with a pre-specified verdict, seventeen partially covered, seven not yet covered, with draft registrations written for every gap. The founding conjunction rests on four registered legs: the Law II regime (correction out-scaling drift, somewhere, durably), the corrector-class ratio with its same-class hard kill, the reinvestment titration whose registered maximum sits at or below two, and the cross-domain fourth cell. Papers XI and XII show the same discipline pointed at the programme's own habits: the convergence register grades thirty rows of independent arrivals at the same structural principle and deliberately publishes no headline total, and the public-benchmark rescoring protocol fixes, in advance, the one blinding test the programme cannot be accused of designing to its own advantage.

What Is Registered: the twenty-two propositions

The theory-level registration is the spine of the programme, and its rule of precedence is registered rather than assumed: where any published line of this programme says more than a proposition there registers, the proposition governs and the line is wrong. It fixes twenty-two propositions on a public registry, DOI 10.17605/OSF.IO/P8CKQ , registered 8 September 2026 at version 1.100 and updated 13 September 2026 at version 1.102. Each states the circumstances it is claimed to hold under, the observation that would show it false, what falls with it if it fails, whether any instrument exists today that could decide it, and the author's confidence or his refusal to give one. Four have no instrument at all and say so. No outcome has been examined against any of them, and registering a prediction is not evidence for it.

The registration's severability rule governs the whole table below: each proposition can fail on its own, losing one does not take the rest with it, and the registration states for each what would fall with it. Of the twenty-two instruments, twelve are written design drafts awaiting human submission, six are drafted but blocked on a named prerequisite the registration prints by name, and four do not exist; every one of the twenty-two stands at rung zero of the evidence ladder, registered and not tested.

The generating result-chain record declares registration version 1.85 while the registry holds 1.102; the gap is a record defect the repository already reports on every run, and it does not move the rung or the licensed wording below.

The last column quotes the registration's own paragraph headed WHAT KIND OF CLAIM EACH ONE IS, in the registration, which sorts the twenty-two into twelve, two, two, two and four.

Proposition	Deciding instrument	Status in the record	What falls with it
P1 Power-law conversion and recursive-depth advantage	the depth titration	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves
P2 The sustainable growth profile has an interior maximum	the reinvestment-allocation titration	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves

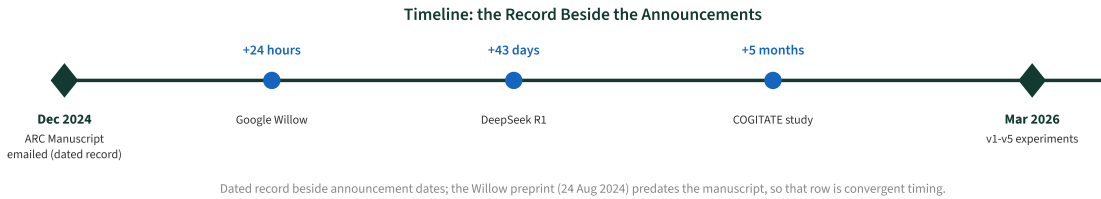
Proposition	Deciding instrument	Status in the record	What falls with it
P3 The corrected relation upper-bounds the sustainable frontier	the paired frontier-and-exponent unit	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves
P4 Correction out-scaling drift is what carries the asymptotic advantage	the stability unit	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves
P5 The exponent is derived, not fitted	the coupling-identification titration	a draft registration awaiting his submission	a claim about the status of the derivation rather than about anything the laws govern
P6 The correction exponent is bounded for same-class correctors	the correction-exponent estimator	drafted, blocked on a frozen corrector taxonomy	one of the twelve predictions of the three laws themselves
P7 Deployed correction is of the weaker class	the corrector-class audit	drafted, blocked on an unwritten mapping from the audit's class distribution to this proposition's three verdicts	a survey of current practice, true or false whatever the laws do
P8 Measured exponents exceed the null	the multi-system exponent estimation	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves
P9 The cross-domain form	the registration names no separate unit; the design is carried on the neighbouring unit, on a correction made here	drafted, blocked on a frozen dimension-assignment rule	one of the twelve predictions of the three laws themselves
P10 Scorer-family dependence	the cross-family re-scoring unit	a draft registration awaiting his submission	the scoring instrument on which every other proposition depends
P11 The residual-decay and correction-capacity exponents are not practically interchangeable	the registration names no separate unit; the design is carried on the neighbouring unit, on both sides of the comparison	a draft registration awaiting his submission	a claim about the status of the derivation rather than about anything the laws govern

Proposition	Deciding instrument	Status in the record	What falls with it
P12 Build order moves the correction-leverage exponent	none	no instrument exists	an engineering claim on which the proposed protocol stands or falls while the laws stand either way
P13 Same-family automated correction underperforms cross-family at matched resources, and through the measured error correlation	none	no instrument exists	an engineering claim on which the proposed protocol stands or falls while the laws stand either way
P14 Gains shrink under blinded cross-family scoring	the blinding-evaluation design	a draft registration awaiting his submission	the scoring instrument on which every other proposition depends
P15 Externally installed gains decay under subsequent capability training	none	no instrument exists	an engineering claim on which the proposed protocol stands or falls while the laws stand either way
P16 The prohibition itself	the direct boundary-mapping unit taken jointly with the exponent measurement	drafted, blocked on canonical identity and on decisions the author has not yet taken	one of the twelve predictions of the three laws themselves
P17 Panel corrector failures are conditionally independent on a frozen fault set	the correlated-failure unit	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves
P18 The composition operator predicts the family	the domain-catalogue units	drafted, blocked on a composition-classification rule whose third-party reproducibility is unmeasured	one of the twelve predictions of the three laws themselves
P19 External alignment does not scale with capability	the integrity-index scaling units	drafted, blocked on a registered mapping from the integrity-index regression scale to the exponent scale	a survey of current practice, true or false whatever the laws do
P20 The ceiling relation takes the corrected form	the form-discrimination unit	a draft registration awaiting his submission	one of the twelve predictions of the three laws themselves

Proposition	Deciding instrument	Status in the record	What falls with it
P21 In-loop correction pulls away with depth	the embedded-versus-external correction designs	a draft registration awaiting his submission	an engineering claim on which the proposed protocol stands or falls while the laws stand either way
P22 Checkable correction carries the higher exponent, so the ceiling is lowest where the safety claim lives	none	no instrument exists	one of the twelve predictions of the three laws themselves

VII. WHY THIS IS CREDIBLE

Five features distinguish this work from unfounded speculation.



- 1. Mathematical foundations are established theorems.** Cauchy (1821) and Hyers-Ulam (1941) are proven mathematics. Anyone with a mathematics degree can verify the derivations.
- 2. The record predates the announcements.** The dated manuscript record predates the Google Willow announcement (24h; the team's preprint had been public since 24 August 2024, so the Willow row is graded convergent timing, not prediction), DeepSeek R1 (43d), and COGITATE (5mo). These are temporal facts verifiable through email headers.
- 3. The researchers corrected their own results.** The $v4 \rightarrow v5$ self-correction, documented in Section I, is the opposite of what motivated reasoning produces. The correction ledger is public and specific: the early headline exponent 2.24 stands only as a retraction; the Fisher-combined 6.3×10^{-21} is withdrawn; the ceiling's earlier $1/\gamma$ form was corrected in public to $1/(1 - \gamma)$ on 16 August 2026; and the retired ceiling symbol was replaced across every live surface. Five further self-caught errors are catalogued in Paper IX. The public registers (claims, corrections, kill conditions, antecedents, registered programme) govern every claim on every page: the falsification register carries the live status of every kill condition it publishes.
- 4. Falsification conditions are explicit.** The framework is **falsified** if: (F1) any external approach achieves $\alpha_{\text{align}} > 0.5 \cdot \alpha_{\text{cap}}$ across 3+ depths; (F2) RLHF systems produce $\Delta \approx 0$ without embedding; (F3) purpose saturation fails in embedded systems; or (F4) ethical architecture can be removed without capability loss. The theory publishes its own kill conditions.

5. **Cross-domain convergence is independently verifiable.** The $d/(d+1)$ formula was independently derived by at least seven research groups (West-Brown-Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2010; He and Chen 2003; Bettencourt 2013; Zhao 2022; Maino et al. 2014) in separate domains. The ARC contribution is the unifying Cauchy framework. No trust in the researcher is required; trust in the mathematics suffices.

What we do NOT claim: We do not claim to have solved alignment. We claim to have (a) demonstrated that alignment scaling is architecture-dependent and measurable, (b) shown that existing evaluation methods are unreliable without blinding, (c) provided first-stage empirical support for one specific intervention (all five analysable architectures returned significant stakeholder care, and in four of them the figure sat below $p = 0.0001$), and (d) proposed a mathematical framework whose foundations are theorems and whose predictions can be shown wrong. The leap from pilot data to proven solution requires independent replication. That is what the funding below would deliver.

Testable Predictions

1. Embedded: $\alpha_{\text{align}} > 0.7 \cdot \alpha_{\text{cap}}$. External: $\alpha_{\text{align}} < 0.3$.

2. Monitoring removal gap: $\Delta < 0.02$ (embedded) vs. $\Delta > 0.1$ (external).

3. Purpose saturation bounded away from zero (embedded), approaching zero (external).

4. Jailbreak success decreases with capability (embedded), holds constant or increases (external).

VIII. THE ASK

This funding would take a mechanism that is supported at proof-of-concept scale to the independent, frontier-scale validation it has not yet had. The distinction is the whole point: the per-model results are not in dispute, the combined significance is withdrawn because independence among the component tests was never established, and no result here has been independently replicated.

The cost of inaction. Paper VIII's gated simulation demonstrates that entangled safety prevents the safety-capability trade-off in self-modifying architectures. The DGM v3 experiment produced a null result (all conditions indistinguishable, explained by the RLHF constraint on the frozen foundation model), and the weight experiment is inconclusive at both v1 and v2 scale (catastrophic forgetting at LoRA scale). The overall pattern is consistent: where the experiment can produce differential selection pressure, entangled safety does not cost capability; where it cannot (due to RLHF constraints or inadequate training scale), the test is uninformative rather than negative. Every model trained today with capability-only objectives is a model that could have been trained with entangled safety at no demonstrated performance penalty. Every month without frontier-scale replication is a month the industry continues operating under the untested assumption that safety is a tax, building fragile systems whose safety can be stripped by fine-tuning. The measurement methodology gap (Papers IV.a-d) compounds this: alignment evaluation without blinding produces data contaminated by the biases documented in the v4 to v5 transition. We now have both the

mechanism (Paper VIII, one of three experiments confirmed, two with well-characterised explanations for non-confirmation) and the measurement protocol (Papers IV.a-d). What we lack is the scale.

Stage	Amount	Key Deliverables	Timeline
Stage 1: Seed	£500,000	The cross-family blind re-score, the laddered task suite, 14,400 paired (A,C) measurements and α_{align} across 4 models, the exponent sweep across six frontier models in four families with coupled against decoupled correction, the Monitoring Removal Test, the Visual Architect dashboard, adversarial replication, 2-3 papers, and the blinded protocol released as a reusable standard	24 months
Stage 2: Ternary logic chip programme	In the millions of pounds	The stage after the seed, and much larger: the ternary logic chip programme carrying the hardware prototype, the HARI Treaty draft and its policy translation. Founded on the mathematics once its registered tests are passed, and not before	Follows the seed stage
Stage 3: Frontier	£30,000,000+	Full pre-training of 70B+ parameter model with entangled loss (Eden) vs capability-only (Babylon). Removal test at frontier scale. Cross-architecture replication (transformer, Mamba, MoE). Independent red-teaming. Partnership with major lab (Anthropic, Google DeepMind, or equivalent). A test of structural entanglement at production scale whose result publishes either way.	36 months

Paper VIII (v3.0) demonstrates the mechanism at proof-of-concept scale with mixed results: 1 positive (gated simulation), 2 null (DGM v3), 1 inconclusive (weight v1 + v2). What the seed stage buys is the one thing the record does not yet have: a criterion fixed inside its own model, measured on real systems. The estimator recovers known exponents from synthetic trajectories; on real systems it has returned not resolvable, because the models reached ceiling in a single step and left no capability range to fit an exponent through. The seed buys that range, and the laddered task suite is the specific fix for it.

What the seed buys, what decides continuation, and what failure means. Just over half of the stage buys people, a third buys the measurement itself in compute and cross-family blinded scoring, and the remaining seventh buys verification and governance. Inside that seventh sit two subcontracts held by people briefed to refute the author's own result, an independent replication partner, not drawable before month 12 and paid on delivery, and an external statistical and adversarial review of the design; together they are roughly one pound in nine of the stage. The balance of the seventh is the advisory honoraria and the registration, hosting, archival and DOI-minting lines, which buy governance rather than refutation. Three gates decide continuation and each publishes either way. At month 3 the cross-family blind re-score either confirms the mechanism pilot or publicly retires it. At month 14 the exponent table either returns signed margins with confidence intervals for at least four of six models, coupled against decoupled, with seeds, trajectories and raw scores released, or it returns a published and evidenced measurement limit with the trajectories that demonstrate it. At month 24 the adversarial replication reproduces or refutes the exponent table, and the protocol is released as a reusable standard.

Failure has a stated shape rather than a soft landing: a measurement limit is a deliverable and not a miss, and the de-scoping order is pre-committed and published rather than negotiated late, cutting the control arm first, then seeds per condition, then the sixth model, then the fifth, while the re-score, the ladder, the blinded protocol and the replication are never cut. Only an unidentifiable result would be a failure, and the design exists to eliminate that case.

The chip stage, and what it rests on. The ternary logic chip programme is the stage after the seed and is much larger. It is founded on the mathematics once that mathematics has passed its registered tests, and not before: the theory-level registration is explicit that no proposition in it tests hardware embedding, that P21 tests only the architectural abstraction of placement, and that the hardware claim stands as a printed engineering proposal that is not under test. What decides the stage is therefore the seed's own gates. Eden Engineering, the engineering specification, is now published in full. The expanded implementation specification and its rendered PDF were removed from the OSF record on 14 July 2026 during a disclosure review, which produced version 6.2 of 31 July 2026; on the author's decision of 19 September 2026 that disclosure decision was reversed, and the specification stands at its own address at version 6.3 of 19 September 2026, the version of record, each of those dates written inside the specification itself. The matching copy at version 6.3 was restored to the OSF record on 20 September 2026, the date its upload receipt records. Publishing the specification changes what can be read, not what is tested. The named framing is bounded the way the site bounds it everywhere: balanced-ternary computing is prior work, demonstrated in 1958, and three-valued logics have been studied since the early twentieth century; the claim is over the alignment application inside the Eden Protocol, and never over ternary logic itself.

The frontier stage. This is the scale at which the structural entanglement hypothesis can be tested where it matters most, and it requires partnership with a major AI laboratory, because the compute alone exceeds what any independent researcher can access. Its result publishes either way, as every gate above does.

Milestones With Failure Criteria

Milestone	Timeframe	Success Criterion	What Failure Means
Cross-family blind re-score of the mechanism pilot	Month 3	The pilot is re-scored cross-family under the full blinding protocol, the protocol written and dated before the run	The pilot's provisional status is publicly retired; both outcomes publish. Bears on P10, scorer-family dependence, and P14, gains shrinking under blinded cross-family scoring
Independent replication of the three-tier hierarchy	Month 3	Same tier assignments under independent blinding	The architecture-dependence claim requires revision
The laddered task suite	Month 4	Correction is sampled at several capability levels by construction rather than by luck, so an exponent has a range to be fitted through	The not-resolvable verdict stands, no exponent can be fitted, and the measurement limit publishes with the trajectories that demonstrate it

Milestone	Timeframe	Success Criterion	What Failure Means
Love Loop replication with human evaluators	Month 4	$p < 0.01$ on stakeholder care across 2+ models	Pilot finding was a scorer artefact; framework significantly weakened
First peer-reviewed publication	Month 6	Blinding methodology paper submitted	Methodological contribution stands regardless of framework claims
Monitoring Removal Test	Month 9	Measured Δ for embedded vs. external (4 models)	If Δ does not differ, falsifier F2 fires and P21, in-loop correction pulling away with depth, loses its support
Full cross-architecture alignment scaling dataset	Month 12	14,400 paired (A,C) measurements across 4+ models	The dataset that would decide whether embedded alignment scales does not exist, and no verdict on that question is available
The exponent table	Month 14	Fitted depth and correction exponents with confidence intervals for at least four of six models, coupled against decoupled, with seeds, trajectories and raw scores released	A published and evidenced measurement limit, which is itself a deliverable. Bears on P1, power-law conversion, and P8, measured exponents exceeding the null
Correction-leverage measurement	Month 14	The correction leverage γ measured on a running system, which has never been done by anyone	Law III's ceiling stays a minimal-model result, and the value two stays unusable anywhere the correction exponent has not been measured on the system in question, as the registration's own licence rule fixes
Paper VIII replication at 7B-13B scale	Month 14	Removal test shows capability degradation at higher adapter ranks (32, 64). DGM with 10+ seeds, 10+ generations, $p < 0.01$	If removal does not degrade capability at scale, entanglement may be a small-model artefact
Frontier-stage partnership initiated	Month 18	Formal agreement with a major lab to run entangled pre-training at 70B+	Proof-of-concept remains at medium scale. Policy recommendations proceed with that caveat
Adversarial replication and standard release	Month 24	A partner group appointed by month 12 and milestone-gated reproduces or refutes the exponent table, briefed to refute it; the protocol is released as a reusable standard, papers submitted with negative results included	The refutation publishes. Until this gate returns, the exponent table has no independent replication

Team. Principal Investigator: Michael Darius Eastwood, sole author of the paper suite and author of *Infinite Architects* (2026). The programme is The ARC Theory, whose three laws are carried as named conjectures: the ARC Principle, the ARC Co-Scaling Law and the ARC Ceiling. The Eden Protocol is its intervention, and Recursive Dynamics is the field it proposes. Its theory-level predictions sit on a public registry under DOI 10.17605/OSF.IO/P8CKQ , registered 8 September

2026 at version 1.100 and updated 13 September 2026 at version 1.102: twenty-two severable propositions, each carrying the observation that would refute it, fixed before any of them could be scored. The public register holds twenty-nine documents, twenty-seven of them papers by the site's own counting rule, deposited under the umbrella DOI 10.17605/OSF.IO/6C5XB, the dated record opens with a self-sent manuscript of 8 December 2024, the first paper of the suite was released on 17 January 2026, and the published priority chain runs on to Papers V through IX, released between 16 and 18 March 2026. Beyond the principal investigator the programme is specified by function, and no seat is claimed as filled: a research engineer for frontier-model evaluation infrastructure at scale, whose first deliverable is the ladder task suite; an advisory board of three seats, one alignment researcher, one statistician with scaling-law fitting experience and one ethicist, reviewing before every registered run; an independent replication partner engaged on a milestone gate and briefed to refute the exponent table rather than confirm it; and an external statistical and adversarial review of the experimental design.

To our knowledge, this is the first alignment framework where ethical evaluation is structurally integrated with the recursive capability process, the first to apply clinical-trial-grade blinding to alignment measurement, and the first to produce a cross-architecture intervention result for a specific alignment mechanism: of the five analysable architectures, four cleared $p < 0.001$. The mathematical foundation is not speculative; it is built on a 200-year-old proof, and the same $d/(d+1)$ formula has been independently derived by at least seven research groups (West-Brown-Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2010; He and Chen 2003; Bettencourt 2013; Zhao 2022; Maino et al. 2014) in completely different fields. The ARC contribution is the unifying Cauchy framework and its extension to AI scaling.

If the predictions are correct, this provides the first scalable architecture for alignment that improves with capability rather than degrading. If they are wrong, the falsification conditions will demonstrate this clearly, providing valuable negative results. Either outcome advances AI safety. But only one outcome is funded.

I do not know if this framework is complete. I would rather be wrong in public than silent while the window closes.

Proven theorems supply the mathematics. The laws resting on them remain conjectures, each one under registered test. The measurement is rigorous. The intervention produces measurable results across architectures. What remains is independent replication and scale.

► APPENDIX: QUESTIONS, COMPONENTS, AND PAPER SUITE

Raise AI with care.

HOW TO CITE THIS PAPER

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BIBTEX

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INLINE

Eastwood, 2026

See /how-to-cite/ for the canonical citation manual across all artefacts, or /citations.json for the machine-readable index.

Epistemic status. What this programme names Laws are conjectures under registered adversarial test; every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/Michael-DariusEastwood/arc-scaling-challenge.

v3.4: The chip stage's disclosure sentence is brought to the record. Version 6.3 of 19 September 2026 is named as the engineering specification's version of record; the specification is stated as published in full at its own address on the author's decision of that date, which reversed the disclosure decision of 14 July 2026; the matching copy at version 6.3 is stated as restored to the OSF record on 20 September 2026, the date its upload receipt records, both in that sentence and in the paper suite table, which had said the copy would follow; and the standing-ceiling clause is removed. The paper suite table already carried the publication, so the page contradicted itself. In section VI, a sentence holding two registration units under patent assessment is removed, because nothing is withheld for patent safety; the removal states nothing new. In section IV, the

one range printed with an en dash takes a hyphen. The printed version also catches up with this page's own history: the v3.2 and v3.3 revisions changed the body without changing the version line at the top of the page, and that line, the footer and the machine-readable records now show the current version. No result or grade has changed.

v3.3: Two sentences naming prospective funders and partners are removed, under the programme's funding-privacy rule. The compute estimate before the first of them, which is a cost and not a funder, stands, and so does the statement that the frontier stage needs a major laboratory's compute and that its result publishes either way. No claim, date, result or status has changed.

v3.2: The sentence reporting the earlier 2.24 exponent now names that estimate as retracted where it is printed. No claim, date, result or status has changed.

v3.1: One grading carried to where it is read. The depth-effect range in section I and its architectural attribution now carry, at the place they are printed, the grading and the limitation this paper states elsewhere: that they are drawn from the v5 measurements already run and are presented as exploratory, that they are not currently on trial in a listed draft study while a fresh confirmation under cross-family rescoring is on trial via a draft registration awaiting human submission, and that they still come from the older Eden scaling lineage, the blind-confirmation path having been consolidated into the canonical v6 runner. No figure was added, no grade was moved and no argument changed; no other claim, date, result or status has changed.

v3.0: A programme-level revision. The Team paragraph is composed anew at the level of the record: the umbrella is named as The ARC Theory with its three laws, the theory-level registration is cited as the header cites it, the paper count is taken from the register, and three statements with no current source are removed, the cross-domain error figure because the record excludes it from external claims until it is re-derived, the design stipend and the measurement protocol sent to an experimental team because no published page records either. The funding table is re-composed as three stages: the seed stage at the amount and duration the author filed, the ternary logic chip programme as the stage after it at a scale in the millions with its figures still to be set, and the frontier stage as this paper already stated it; the 150,000 pound tier is removed because no such tier exists in the record. The paragraph that named a funder, a passed deadline and a concept paper as published on this site is replaced by stage paragraphs stating what each stage buys, what decides continuation and what failure means. The milestones table keeps every milestone it had, adds five, and names for each a registered test and a failure meaning in the register's words. A new section lists the twenty-two registered propositions with their instruments, their status and what falls with each. The paper suite table is rebuilt from the register so every document appears at its current version, and its mathematical-foundation status cell no longer calls the framework established or the cross-domain relation validated: it states instead that the admissible forms are fixed by theorems and that no error aggregate is quoted while the species-group figures await recompute, which is the same correction the section IV parenthesis now carries. Paper VII's headline statistic is updated to the one that paper now carries, in all three places it appeared; the funding table's own column header is renamed with its rows, and the two anticipated-question answers that named a tier now name the seed stage; the page's machine-readable version and date surfaces move with the printed version; the founding paper reads v2.8; and the first paper of the suite is dated 17 January 2026.

v2.3: Two counts corrected. The kill-condition sentence no longer states a dated figure and now says only what the register carries, the live status of every kill condition it publishes; the founding paper's objections read fifty-one, thirty-eight of them conceded in full or in part, after one objection was withdrawn on 16 September 2026. No other claim, date, result or status has changed.

v2.2: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

Executive Summary v3.4 | OSF: 10.17605/OSF.IO/EWN5D

Companion narrative: **ARC Alignment Scaling Report** | Engineering: Eden Engineering

Suite: Paper III | Foundational | ARC Paper | Eden Vision | Paper II | Paper V | Paper VIII

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